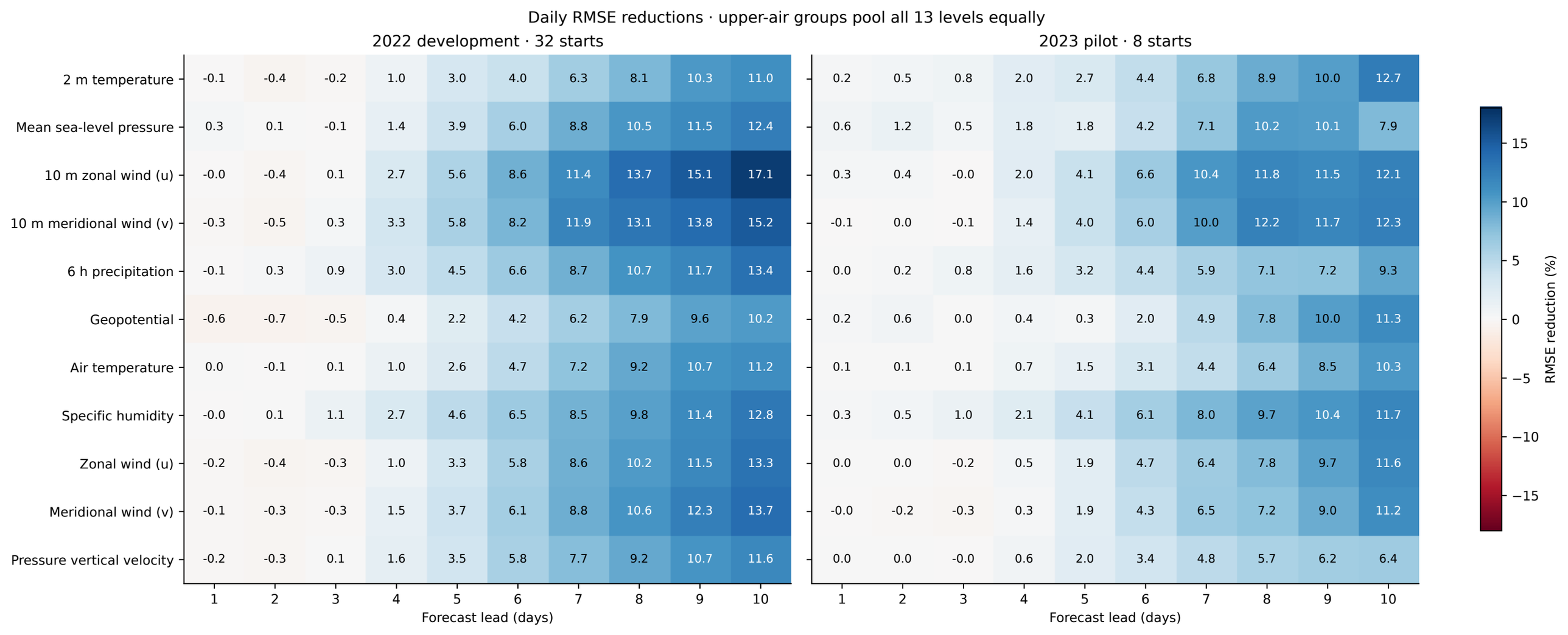
Daily RMSE reductions · upper-air groups pool all 13 levels equally
2022 development · 32 starts
2023 pilot · 8 starts
2 m temperature
Mean sea-level pressure
10 m zonal wind (u)
10 m meridional wind (v)
6 h precipitation
Geopotential
Air temperature
Specific humidity
Zonal wind (u)
Meridional wind (v)
Pressure vertical velocity
-0.1 -0.4 -0.2 1.0 3.0 4.0 6.3 8.1 10.3 11.0
0.3 0.1 -0.1 1.4 3.9 6.0 8.8 10.5 11.5 12.4
-0.0 -0.4 0.1 2.7 5.6 8.6 11.4 13.7 15.1 17.1
-0.3 -0.5 0.3 3.3 5.8 8.2 11.9 13.1 13.8 15.2
-0.1 0.3 0.9 3.0 4.5 6.6 8.7 10.7 11.7 13.4
-0.6 -0.7 -0.5 0.4 2.2 4.2 6.2 7.9 9.6 10.2
0.0 -0.1 0.1 1.0 2.6 4.7 7.2 9.2 10.7 11.2
-0.0 0.1 1.1 2.7 4.6 6.5 8.5 9.8 11.4 12.8
-0.2 -0.4 -0.3 1.0 3.3 5.8 8.6 10.2 11.5 13.3
-0.1 -0.3 -0.3 1.5 3.7 6.1 8.8 10.6 12.3 13.7
-0.2 -0.3 0.1 1.6 3.5 5.8 7.7 9.2 10.7 11.6
0.2 0.5 0.8 2.0 2.7 4.4 6.8 8.9 10.0 12.7
0.6 1.2 0.5 1.8 1.8 4.2 7.1 10.2 10.1 7.9
0.3 0.4 -0.0 2.0 4.1 6.6 10.4 11.8 11.5 12.1
-0.1 0.0 -0.1 1.4 4.0 6.0 10.0 12.2 11.7 12.3
0.0 0.2 0.8 1.6 3.2 4.4 5.9 7.1 7.2 9.3
0.2 0.6 0.0 0.4 0.3 2.0 4.9 7.8 10.0 11.3
0.1 0.1 0.1 0.7 1.5 3.1 4.4 6.4 8.5 10.3
0.3 0.5 1.0 2.1 4.1 6.1 8.0 9.7 10.4 11.7
0.0 0.0 -0.2 0.5 1.9 4.7 6.4 7.8 9.7 11.6
-0.0 -0.2 -0.3 0.3 1.9 4.3 6.5 7.2 9.0 11.2
0.0 0.0 -0.0 0.6 2.0 3.4 4.8 5.7 6.2 6.4
1 2 3 4 5 6 7 8 9 10
Forecast lead (days)
1 2 3 4 5 6 7 8 9 10
Forecast lead (days)
15
10
5
0
−5
−10
−15
RMSE reduction (%)